

Detailed Proposal

Title of Project: Structure-Preserving Conditional Diffusion Models for High-Fidelity Data Synthesis

Summary of the Project

Data-driven modelling has become central to scientific research, engineering analysis, medical decision-making, industrial monitoring, financial modelling, and environmental studies. However, many important data regimes are difficult to observe in sufficient quantity. Rare medical conditions, low-frequency system failures, extreme environmental events, high-risk financial states, sensitive population groups, and costly scientific experiments are all examples where the available data may be limited, imbalanced, or incomplete. Synthetic data generation is often used to address this limitation. However, a major theoretical issue remains. In structured data, the meaning of a sample does not come only from its individual feature values. It also comes from the relationships among variables. A medical state may be expressed through the joint behaviour of several biomarkers; an industrial fault may appear through correlated sensor patterns; an environmental extreme may depend on the interaction of temperature, pressure, humidity, and wind. Therefore, synthetic data that reproduce individual feature ranges but distort inter-variable structure may look plausible while remaining mathematically and scientifically unreliable. This proposal develops a mathematical framework for **structure-preserving conditional data synthesis**. The aim is to generate data for a specific condition without disturbing the main structural properties of the original distribution. Thus, synthetic data should not be treated merely as additional samples. It should be seen as a conditional approximation of the target distribution, where the important relationships in the data are retained.

Let $P_c = P(X \mid C = c)$ denote the true conditional distribution for a condition, class, regime, subgroup, or scientific state c . The task is to learn a generated conditional distribution $\hat{P}_{\theta,c}$ that stays close to P_c :

$$\hat{P}_{\theta,c} \approx P_c. \quad (1)$$

Closeness to P_c alone is not sufficient. The generated distribution should also preserve the structural features of the real data. We represent these features through a family of distributional functionals $\mathcal{S}$. Hence, the generated distribution is also expected to satisfy

$$\mathcal{S}(\hat{P}_{\theta,c}) \approx \mathcal{S}(P_c). \quad (2)$$

This work addresses the problem by combining conditional diffusion modelling, stochastic transport from a Gaussian reference distribution, feature-token operator representations, and structural fidelity measures based on covariance, correlation, and projected distributional geometry. The resulting formulation provides a mathematical basis for generating synthetic data when real observations are limited, sensitive, costly to collect, or governed by strong structural dependencies.

1. Origin of the Proposal

1.1 Mathematical Rationale and Problem Setting

The origin of this proposal lies in a simple observation: real-world structured data are not merely collections of independent numerical values. They are samples from probability distributions with internal organization. This organization appears through correlations, co-variations, conditional dependencies, latent geometry, and regime-specific structure. A synthesis model that ignores these relationships may generate additional samples, but those samples may not preserve the scientific content of the original data.

Let

$$\mathcal{D} = \{(x_i, c_i)\}_{i=1}^n \quad (3)$$

be a structured dataset, where $x_i \in \mathbb{R}^d$ denotes a sample with d variables and $c_i \in \mathcal{C}$ denotes a condition or regime. For a fixed condition c , the target object is the conditional probability measure

$$P_c = P(X \mid C = c). \quad (4)$$

A conventional synthesis model attempts to generate samples whose empirical distribution is close to P_c . This is necessary, but not sufficient for structured data. The generated measure must also preserve structural information. Therefore, the mathematical object of interest is not only the measure P_c , but also a collection of functionals

$$\mathcal{S}(P_c) = \{\Sigma(P_c), R(P_c), \Pi(P_c), \dots\}, \quad (5)$$

where $\Sigma(P_c)$ represents covariance structure, $R(P_c)$ represents correlation structure, and $\Pi(P_c)$ represents projected distributional geometry.

The proposed research studies conditional data synthesis as the problem of constructing a stochastic map

$$T_{\theta,c} : \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad (6)$$

which transports a reference distribution, usually a standard Gaussian distribution $\gamma = \mathcal{N}(0, I)$, to a generated conditional distribution

$$\hat{P}_{\theta,c} = (T_{\theta,c})_{\#}\gamma, \quad (7)$$

where $(T_{\theta,c})_{\#}\gamma$ denotes the pushforward measure of γ under $T_{\theta,c}$. The mathematical goal is to make $\hat{P}_{\theta,c}$ close to P_c , while preserving the structural functionals of P_c .

1.2 Research Gap

The main challenge is not simply the shortage of synthetic data. The more critical issue is the shortage of synthetic data that preserve the structural relationships present in the original distribution. Generative models, including adversarial models, variational models, and diffusion models, have shown strong capability in producing realistic samples [3, 4, 5, 6]. However, for structured-data problems, samples that appear realistic, or even match marginal distributions well, may still fail to preserve the deeper dependency structure of the data.

This gap appears in four important ways:

1. Marginal realism without dependency preservation.

Generated samples may reproduce the distributions of individual variables, but still miss the relationships among them. In such cases, covariance patterns, correlations, and conditional dependencies may become weakened, distorted, or entirely lost during generation.

2. Weak representation of rare conditional measures.

For rare conditions c , the conditional distribution P_c is often estimated from very few observations. This makes the empirical distribution unstable and makes it harder to learn a reliable generated conditional distribution $\hat{P}_{\theta,c}$.

3. Absence of explicit structural constraints.

Many synthetic data generation methods are mainly driven by reconstruction losses, adversarial objectives, or likelihood-based training criteria. These objectives can improve the apparent realism of generated samples, but they do not necessarily enforce the preservation of important structural properties in the generated distribution.

4. Limited geometric evaluation.

Synthetic data is usually evaluated through downstream performance, reconstruction error, or visual inspection. These checks are useful, but they do not tell us enough about whether the shape and geometry of the original distribution have been preserved. A generated dataset may therefore look acceptable under standard tests, yet still fail to capture the structural organization of the real data.

These limitations show the need for a synthesis framework in which structural fidelity is not treated as a secondary concern. It should be built directly into the generation process as a central objective.

1.3 Central Mathematical Aim

The central aim of this proposal is to formulate and study a structure-preserving conditional synthesis model. The problem is expressed as constrained conditional measure approximation:

$$\min_{\theta} \mathcal{D}_0(P_c, \hat{P}_{\theta,c}) + \lambda \mathcal{D}_S(\mathcal{S}(P_c), \mathcal{S}(\hat{P}_{\theta,c})), \quad (8)$$

where $\mathcal{D}_0$ measures distributional discrepancy and $\mathcal{D}_S$ measures discrepancy between structural functionals. The first term controls distributional approximation, while the second term controls structural preservation.

The proposed work therefore studies synthetic data generation as a mathematical problem in conditional probability, stochastic transport, structured operator learning, and variational optimization.

2. State-of-the-Art Review in the Proposed Subject

2.1 Status and Significance of the Proposed Work

Generative modelling has advanced considerably during the last decade. Generative Adversarial Networks introduced data generation as a minimax game between a generator and a discriminator [3]. Variational Autoencoders approached generation through latent-variable modelling and variational inference [4]. Diffusion models further shifted the field by using a forward noise process and learning the reverse denoising dynamics [5, 6]. More recent studies have also adapted diffusion-based methods beyond images, including tabular and other structured-data settings [7].

Although these developments are important, much of the existing work still focuses mainly on distributional approximation, sample realism, or likelihood-based learning. Structured scientific datasets require a more careful treatment. Their value does not depend only on the behaviour of individual variables, but also on the relationships among them, such as dependencies, associations, and interaction patterns. Therefore, a synthetic dataset may look realistic at the level of individual features, but may still fail to preserve the structural relationships that make the original data meaningful for scientific analysis.

The proposed research addresses this issue by placing structural preservation within the mathematical formulation of the generative process itself. Its significance lies in moving beyond conventional sample generation toward structure-preserving conditional measure learning. In this setting, the question is not only whether the generated conditional measure $\hat{P}_{\theta,c}$ approximates the target conditional measure P_c . It is also necessary to examine whether the structural functionals of $\hat{P}_{\theta,c}$ remain consistent with those of P_c . This provides a stronger foundation for synthetic data generation in structured domains, where preserving relationships is as important as reproducing distributional patterns.

2.2 Limitations in Current Research

Several mathematical and methodological limitations remain open in the current literature:

1. Most generative objectives do not directly enforce the preservation of second-order structural properties, such as covariance, correlation, and related dependency measures.
2. Conditional generation is still difficult in rare regimes. In such cases, the target conditional measure P_c is often estimated from only a few observations. The estimate therefore becomes unstable, and the generated measure learned from it may not be reliable.
3. Many existing methods represent structured data as ordinary feature vectors. This simplification can miss higher-order interactions among variables, particularly when the data contain complex dependency patterns.
4. Distributional fidelity and structural fidelity are usually treated separately. There is still a need to bring them together within a single variational or transport-based formulation.
5. Projected geometric discrepancies, such as sliced Wasserstein-type distances, are not yet used sufficiently in conditional synthesis methods for structured data.

The proposed framework addresses these issues by combining conditional diffusion transport with structural fidelity functionals. In this way, distributional accuracy and structural preservation are handled together within one mathematical formulation for reliable synthetic data generation.

3. Proposed Mathematical Framework

3.1 Conditional Measure Formulation

For each condition $c \in \mathcal{C}$, let P_c be the target conditional measure on $\mathbb{R}^d$. The synthesis model defines a generated conditional measure $\hat{P}_{\theta,c}$. The basic approximation requirement is

$$\hat{P}_{\theta,c} \approx P_c. \tag{9}$$

However, for structured data, this is strengthened to

$$\left(\hat{P}_{\theta,c}, \mathcal{S}(\hat{P}_{\theta,c})\right) \approx (P_c, \mathcal{S}(P_c)). \quad (10)$$

This formulation treats a distribution together with its structural descriptors. The generated measure is acceptable only when both its probability mass and its structural properties align with the target conditional measure.

3.2 Diffusion as Conditional Stochastic Transport

The diffusion component is viewed as a conditional stochastic transport process. Starting from a clean standardized sample $\bar{x}$, the forward corruption process is

$$\bar{x}_\sigma = \bar{x} + \sigma\xi, \quad \xi \sim \mathcal{N}(0, I), \quad (11)$$

where $\sigma > 0$ denotes the noise level. The reverse process learns a denoising map

$$D_\theta(\bar{x}_\sigma, \sigma, c), \quad (12)$$

which estimates $\bar{x}$ from $\bar{x}_\sigma$ under condition c . The denoising objective is

$$\mathcal{L}_{diff} = \mathbb{E}_{\bar{x}, c, \sigma, \xi} \left[w(\sigma) \|D_\theta(\bar{x}_\sigma, \sigma, c) - \bar{x}\|_2^2 \right]. \quad (13)$$

From a transport perspective, the learned reverse process constructs a map from a reference Gaussian measure to a target conditional measure. If $z \sim \mathcal{N}(0, I)$, then the generated sample is represented as

$$\tilde{x} = T_{\theta,c}(z), \quad (14)$$

and the induced generated measure is

$$\hat{P}_{\theta,c} = (T_{\theta,c})_\# \mathcal{N}(0, I). \quad (15)$$

Thus, conditional diffusion is interpreted as a stochastic mechanism for approximating the target measure P_c .

3.3 Feature-Token Representation as a Structured Operator

A structured sample $x \in \mathbb{R}^d$ contains d variables. Instead of treating x only as a flat vector, the proposed mathematical model represents it as a sequence of variable-level tokens:

$$T(x) = \{t_1, t_2, \dots, t_d\}. \quad (16)$$

For a noisy standardized sample $\bar{x}_\sigma$, each token is defined as

$$t_j = \phi_{tok}(\bar{x}_{\sigma,j}) + e_j + \phi_c(c) + \phi_\sigma(\sigma), \quad (17)$$

where e_j is the feature-identity embedding, $\phi_c(c)$ is the condition embedding, and $\phi_\sigma(\sigma)$ is the noise-level embedding. This representation allows the denoising model to act as an operator on variables rather than only as a function on an unstructured vector.

Let $\mathcal{A}_\theta$ denote the token-interaction operator. A layer-wise representation can be written as

$$H^{(\ell+1)} = \mathcal{A}_\theta^{(\ell)} \left(H^{(\ell)}, c, \sigma \right), \quad \ell = 0, \dots, L-1. \quad (18)$$

This operator-based view is useful because structural data require the model to learn relations among variables. The role of the token operator is to capture such relations during the denoising process.

3.4 Structural Fidelity Functionals

The central mathematical contribution of the proposal is the use of structural fidelity functionals. Let X_c denote samples from the real conditional distribution and $\widehat{X}_c$ denote samples from the generated conditional distribution. The structural discrepancy is expressed through complementary functionals.

3.4.1. Covariance Functional.. Covariance captures scale-dependent co-variation among variables. The covariance discrepancy is

$$\mathcal{L}_{cov} = \frac{1}{d} \left\| \text{Cov}(\widehat{X}_c) - \text{Cov}(X_c) \right\|_F. \quad (19)$$

3.4.2. Correlation Functional.. Correlation captures normalized dependence and reduces sensitivity to scale. The correlation discrepancy is

$$\mathcal{L}_{corr} = \frac{1}{d} \left\| \text{Corr}(\widehat{X}_c) - \text{Corr}(X_c) \right\|_F. \quad (20)$$

3.4.3. Projected Geometry Functional.. Second-order structure does not fully characterize a distribution. Therefore, projected geometry is measured through sliced Wasserstein distance. Let $v_k \in \mathbb{S}^{d-1}$ be a random unit projection direction. The projected discrepancy is

$$\mathcal{L}_{sw} = \frac{1}{K} \sum_{k=1}^K W_1 \left(\langle \widehat{X}_c, v_k \rangle, \langle X_c, v_k \rangle \right), \quad (21)$$

where W_1 is the one-dimensional Wasserstein distance. This term compares the generated and real conditional distributions across multiple projected views.

Together, these functionals define

$$\mathcal{D}_S = \lambda_{cov} \mathcal{L}_{cov} + \lambda_{corr} \mathcal{L}_{corr} + \lambda_{sw} \mathcal{L}_{sw}. \quad (22)$$

3.5 Unified Variational Objective

The complete mathematical objective combines diffusion-based conditional generation with structural preservation:

$$\mathcal{L} = \mathcal{L}_{diff} + \lambda_{cov} \mathcal{L}_{cov} + \lambda_{corr} \mathcal{L}_{corr} + \lambda_{sw} \mathcal{L}_{sw}. \quad (23)$$

This objective has two roles. The first term learns the reverse stochastic transport needed to generate condition-specific samples. The remaining terms constrain the generated measure so that it preserves the structural properties of the target conditional measure.

Equivalently, the proposed framework can be interpreted as the following constrained optimization problem:

$$\min_{\theta} \mathcal{L}_{diff}(\theta) \quad \text{subject to} \quad \mathcal{S}(\widehat{P}_{\theta,c}) \approx \mathcal{S}(P_c). \quad (24)$$

Using a Lagrangian relaxation gives the unified objective above. This formulation gives a clear mathematical meaning to structure-preserving synthesis.

3.6 Applied Outcome: Suggested Plan of Action for Utilization of Research Outcome

This project has been developed with practical use in mind, especially for situations where sufficient real-world data are not easily available. Such situations may occur because of restricted access, privacy concerns, high data-collection costs, incomplete records, or institutional and regulatory limitations. In such situations, synthetic data should do more than increase the sample size. They should retain the relationships and structural patterns that make the original data useful for scientific analysis.

This work aims to generate condition-specific synthetic data while preserving the main structure of the target distribution. The planned outcomes are listed below.

1. Open-source software toolkit.

A reusable toolkit will be developed for structure-preserving conditional data synthesis. It will include the model implementation, data generation scripts, evaluation code, configuration files, and documentation. This will make the work reproducible and allow researchers to apply or adapt the method to other datasets.

2. Benchmarking and evaluation protocols.

The project will provide practical protocols for evaluating synthetic data. The evaluation will cover statistical similarity, structural similarity, conditional consistency, task-level usefulness, robustness, and computational cost. This is needed because synthetic data may look realistic or perform well in one task, yet still fail to preserve the relationships present in the original data.

3. Applications in scientific and engineering domains.

The method can be useful in fields where real data are difficult to collect, expensive to obtain, sensitive to share, or incomplete. Examples include healthcare, climate studies, industrial monitoring, risk analysis, materials science, and other structured-data problems. In these areas, data are valuable not only because of individual variables, but also because of the relationships among them.

4. Privacy-aware data sharing.

Many real datasets cannot be shared openly because of privacy, legal, ethical, or institutional restrictions. Structurally faithful synthetic data can support collaboration, preliminary analysis, model testing, and method development, while reducing direct exposure of sensitive records.

5. Academic contribution.

The project will contribute to research on conditional generative modelling, structure-preserving synthetic data, synthetic data evaluation, and distributional fidelity analysis. It may also support related work in artificial intelligence, statistics, applied mathematics, and data-driven scientific modelling.

The main outcome is therefore not merely a method for producing more data. The goal is to generate synthetic data that remain analytically useful because they preserve the structural information required for modelling, interpretation, and decision-making.

4. References

References

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